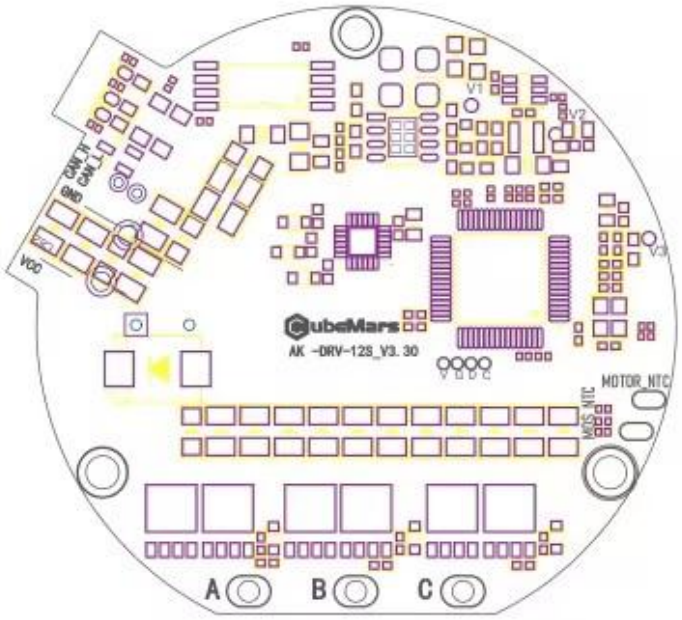
AK60-4820-2D-A2 Driver Manual

V1.0.0



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Precautions

1.Ensure that there are no short circuits in the circuit and that interfaces are connected correctly as required.

2.  The driver board will heat up during output; please use it carefully to avoid burns.

3.  Before use, please check if all parts are intact. If any parts are missing or aged, please stop using it and contact technical support in time.

4.  Please strictly follow the working voltage, current, temperature, and other parameters specified in this document; otherwise, it will cause permanent damage to the product!

Product Features

The AK series motor driver board adopts a mid-to-high-end performance driver chip within its class, utilizing the Field Oriented Control (FOC) algorithm together with advanced Active Disturbance Rejection Control (ADRC) technology for speed and position control. It supports parameter configuration and firmware upgrades through the CubeMarsTool tuning software.

On the hardware side, the inner loop uses a 16-bit high-precision encoder, with support for up to 21-bit resolution (custom firmware required). The CAN communication interface adopts a safer isolated CAN design and upgraded high-reliability connectors, significantly improving product usability and communication reliability.

On the software side, the CubeMarsTool upper-computer software has been completely upgraded. Switching between servo mode and torque control mode is no longer required, the control interface is more streamlined, and many operational steps have been simplified, comprehensively enhancing the overall user experience.

Disclaimer

Thank you for purchasing the AK series driver board. Before using it, please read this statement carefully. Once used, it is considered as recognition and acceptance of all the contents of this statement. Please strictly follow the product manual and relevant laws, regulations, policies, and guidelines for the installation and use of the product. During the use of the product, the user promises to be responsible for their own actions and all consequences arising therefrom.

Any losses caused by improper use, installation, or modification of the product by the user, CubeMars will not assume legal responsibility.

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Version Change Record

Date	Version	Content
2026.6.17	V1.0.0	1.First time edit

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1. Driver Information

Name Rule

AK 80 - 48 30 - 1 A - A 4

Lable	Compatible Type
AK	Module

Lable	Driver Board Outer Diameter
80	80mm

Lable	Rated Working Voltage
48	24：24V
	48：48V

Lable	Rated Ouput Current
30	10：10A
	20：20A
	30：30A

Lable	Number of Onboard Encoders
1	1：Single-encoder
	2：Dual-encoders

Lable	Interface Type
4	1：XT30(2P)+GH125(4P)
	2：XT30(2+2)
	3：XT30(2+2)* 2
	4：XT60PW-M + GH125(4P)
	5：XT60PW-M + XT60PW-F + HY2.0 2P * 2
	6：2+5pin

Lable	Communication Method
A	A：CAN
	B：CAN FD
	C：RS485

Lable	Encoder Mounting Method
A	A：Coaxial mounting (shaft-to-shaft)
	B：Off-axis mounting
	C：Coaxial mounting + external mountable option
	D：Coaxial mounting + coaxial mounting
	E：Coaxial mounting + off-axis mounting
	F：Off-axis mounting + off-axis mounting

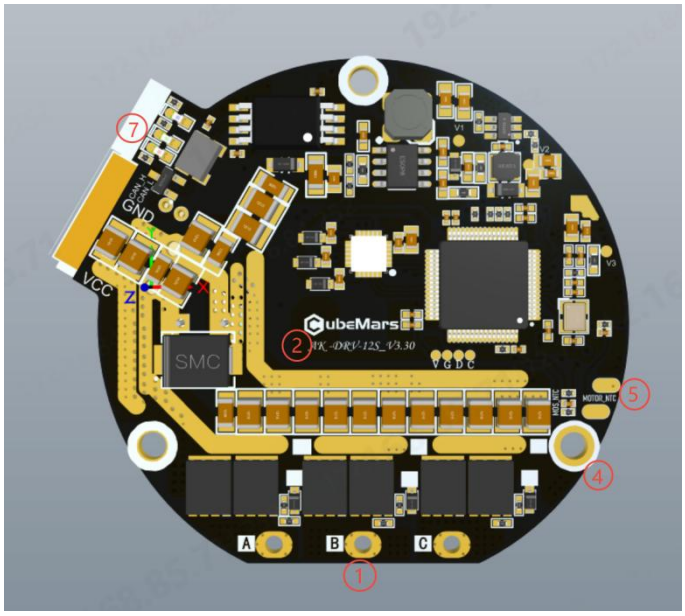
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1.1 Driver Specifications

Model	AK60-4820-2D-A2
Hardware version	AK-DRV-12S_V3.30
Adapted actuator	AK70-9 dual encoders
Rated input voltage	48V
Allowable working voltage	18-52V
Rated output current（rms）	20A
Maximum output current（AP）	60A
Standby power consumption	≤1W
CAN bus bit rate	1Mbps
Size	63mm×57mm
Working environment temperature	-20℃ to 65℃
Maximum allowable temperature for driver board	100℃
Inner Encoder bits	16bit（single turn absolute）
Outer Encoder bits（dual encoder）	16bit（single turn absolute）

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1.2 Appearance Overview



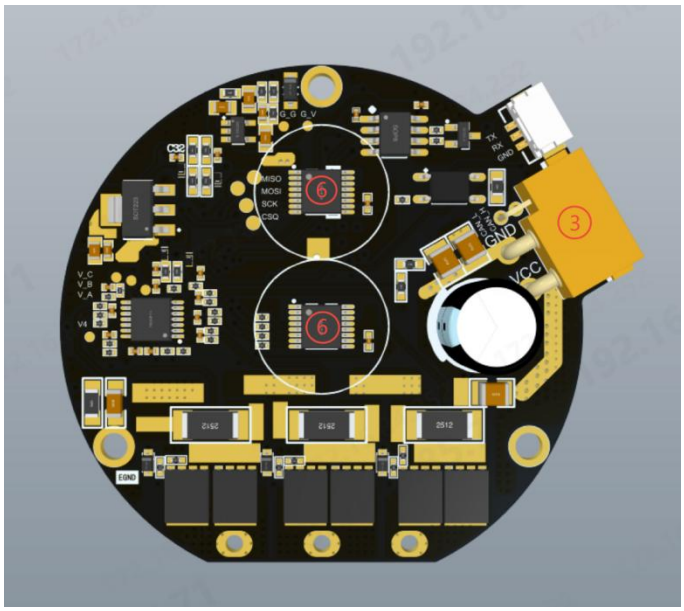
①Three-phase power connection terminal

②Hardware version

③Connection Port

④Mounting hole

⑤Temperature Sensor

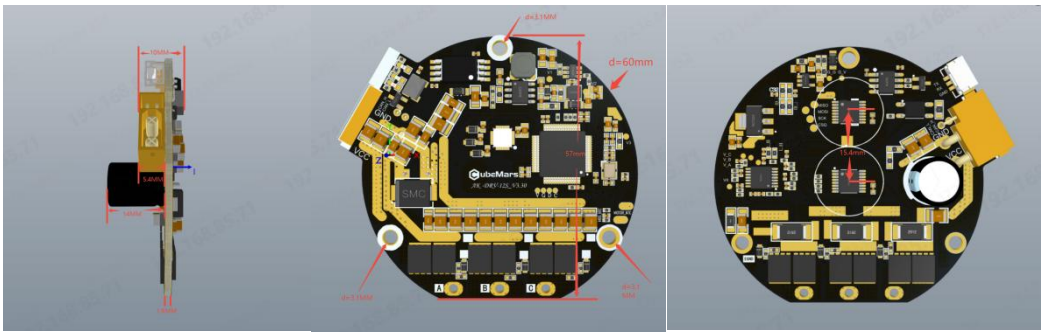


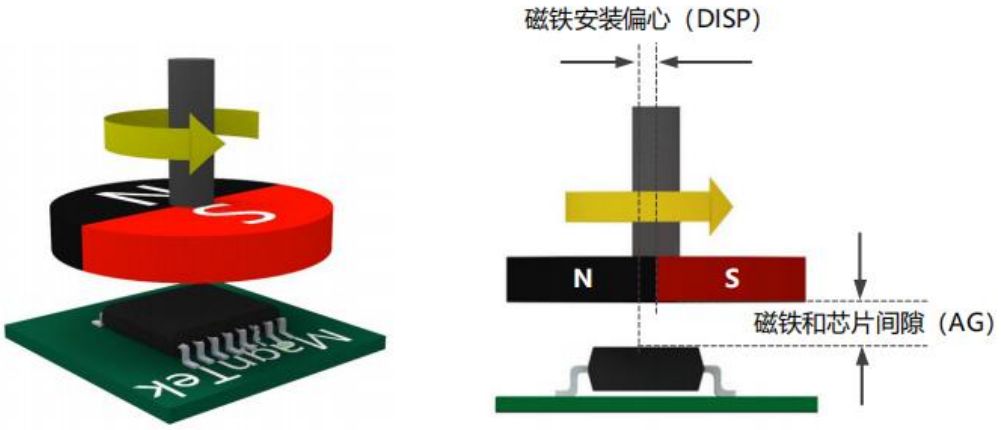
⑥Encoder Chips

⑦LED Indicator

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1.3 Product Dimensions and Installation Notes





Remark：

1、 Dimensional tolerance: ±0.15 mm; mounting hole tolerance: +0.1 mm / -0 mm; PCB thickness tolerance: ±0.16 mm.

2、It is recommended to use M3 mm screws for mounting.;

3、The suggested air gap (AG) between the encoder magnet and the chip is 1 mm, with a maximum not exceeding 3 mm.

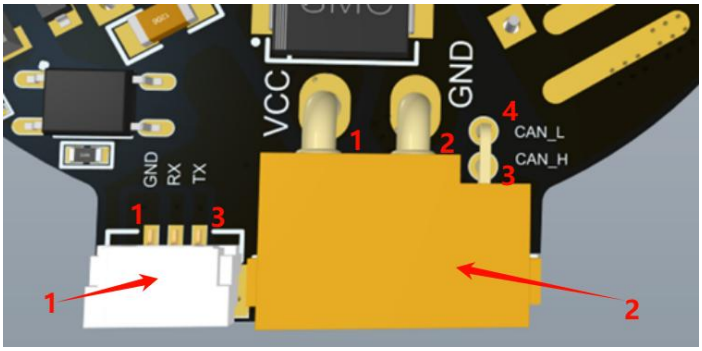
4、The recommended maximum eccentricity (DISP: deviation between magnet center and chip center) is 0.3 mm.

5.See the accompanying drawings for detailed installation instructions.

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1.4 Interface and Definitions

1.4.1 Driver Interface Diagram



1.4.2 Driver Interface Pin Definitions

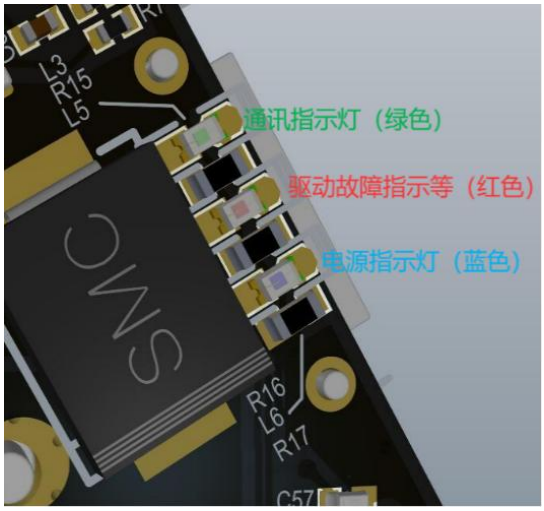
No.	Function	pin	Clarifications	Color
1	**Serial communication**	1	Serial signal ground（GND）	Black
		2	Serial signal input（RX）	Yellow
		3	Serial signal output（TX）	Green
2	**Power input & CAN communication**	1	Positive Pole（+）	Red
		2	Negative Pole（-）	Black
		3	CAN communication high side（CAN_H）	White
		4	CAN communication low side（CAN_L）	Blue

1.4.3 Recommended Brands and Models for Driver Interface

No.	Onboard Interface model	Brand	Terminal Interface Model	Brand
1	A1257WR-S-3P	CJT	A1257H-3P	CJT
2	XT30PW(2+2)-M	AMASS	XT30(2+2)-F	AMASS

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1.4 Indicator Light Definitions



Indicator Light Definitions		
Power Indicator Light（Blue）	Light on	The driver board is powered
	Light off	The driver board is not powered
Operation Indicator Light(Green)	Light on	The motor is working
	Light off	The motor is not working
Drive Fault Indicator Light (Red)	Light on	Driver board fault
	Light off	Driver board function normally

Remark：

⚠：After the driver board is powered, the blue light should remain on in the normal state, and the green and red lights should light up for 2 seconds before going out.

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1.5 Main Accessories and Specifications

No.	Item	Specifications		Quantity	Remark
1	Power and signal cable plug	Power & CAN cable	16 AWG Red/Black Silicone Wire and White/Blue PTFE Wire 30 AWG (OD 0.64 mm), 100 ± 10 mm, 4 wires, XT30(2+2)-F One end is terminated with an XT30(2+2)-F connector, and the other end is stripped and tinned, with a tinned length of 3 ± 1 mm.	Each 1PCS	±2MM
2		Serial cable	30 AWG PTFE Wire (OD 0.64 mm), Length 200 ± 10 mm, GH1.25 3-Pin Male Connector to FC Crimp Housing Header Connector (2×4 Pin)	Each 1PCS	±2MM

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2.Connection

2.1 R-link Appearance Overview, Product Specifications & Interface Definitions



Specifications	
Rated working voltage	5V
VCC Voltage Selection Switch	3.3V（default）
Standby consumption	≤30mA
Dimension	73.8x23.6x14.5MM
Working environment temperature	-20℃ to 65℃
Maximum allowable temperature for controller	85℃

No.	Function	Pin	Clarifications
1	Serial Interface	RXD	Serial Signal Input
		TXD	Serial Signal Output
		GND	Serial Signal Ground
		VCC	VCC Power Output
		RTS	Send Data Request
		CTS	Clear Send
		GND	Power Ground

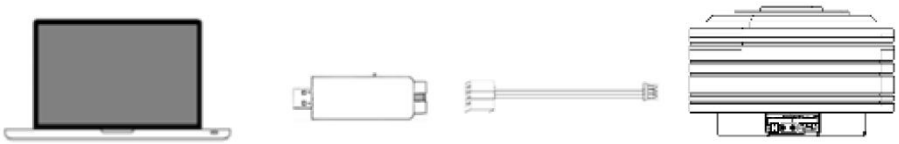
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No.	Function	Pin	Clarifications
		5V	5V Power output

2.2 R-link Indicator Light Definition

No	Light	Color	Clarifications
1	PWR	Red	**Power indicator LED** Red light on when powered. If not lit, disconnect power immediately.
2	TXD	Red	**Serial communication TX indicator LED** Normally off; flashes when data is transmitted.
3	RXD	Red	**Serial communication RX indicator LED** Normally off; flashes when data is received.

2.3Connection Notes



R-Link - USB ----> PC
8Pin interface ----> RLink8P ininterface
3Pin （UART） ----> Motor-3pin（UART）

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